

Grading Rubric for Final Project Presentation

Overview

The final group presentation is your opportunity to showcase your machine learning project, summarize your technical approach, and communicate your findings to a professional audience. Each team will deliver a 10-minute presentation, followed by 2–3 minutes of Q&A.

Your presentation should be clear, concise, engaging, and technically rigorous, demonstrating both your understanding of machine learning methods and your ability to explain your solution to mixed technical and non-technical stakeholders.

Presentation Requirements

1. Duration

- 10 minutes total
- 2–3 minutes Q&A
- Presentations may be stopped at the 10-minute mark.

2. Presentation Content

Your presentation must include the following components:

1. *Title Slide*

- Project title
- Team member names
- Course, semester, and instructor
- Brief one-sentence summary of your project

2. *Problem Definition & Significance (1–2 minutes)*

- What problem are you solving?
- Why is this problem important in the real world?
- Who benefits from solving it?
- What gap in industry, research, or current tools does your project address?

3. Objectives (1 minute)

State your project objectives clearly:

- ML task type (classification, regression, forecasting, detection, clustering, etc.)
- SMART goals
- Expected outcomes (accuracy targets, performance improvements, insights, etc.)

4. Data Source & Preprocessing (2 minutes)

- Dataset origin (Kaggle, UCI, API, IoT sensor, etc.)
- Dataset size, features, and structure
- Preprocessing steps (cleaning, normalization, feature engineering, labeling, balancing)
- Any challenges encountered (missing values, imbalance, noise)

5. Methods & Models (2–3 minutes)

Explain the algorithms and ML pipeline used:

- Models tested (e.g., Random Forest, SVM, KNN, Neural Networks)
- Model selection process
- Hyperparameter tuning (Grid Search, CV, Bayesian Optimization, etc.)
- Performance metrics (accuracy, RMSE, F1-score, AUROC, confusion matrix)

Use 1–2 figures or charts to illustrate key results (e.g., model comparison plots, ROC curves, error bars).

6. Results & Insights (1–2 minutes)

Present your final model's performance:

- Best model and justification
- Strengths and weaknesses
- Misclassification patterns or error analysis
- Visualizations (charts, plots, tables)

7. Impact, Limitations, and Future Work (1 minute)

Discuss:

- Potential real-world application
- Limitations of your current solution
- What improvements would you pursue with more time
- Opportunities for deployment or scaling

8. Conclusion (30 seconds)

Summarize:

- The problem
- Your ML solution
- Key results
- Final takeaways

Presentation Rubric (100 Points Total)

A. Problem Definition & Significance (15 points)

Level	Criteria
Excellent (13–15)	The problem is clearly defined, compelling in significance, strongly real-world relevant, and clearly articulates who benefits and the existing gaps.
Proficient (10–12)	Problem defined with adequate significance; some details lacking depth.
Developing (7–9)	The problem is somewhat vague, or the significance is unclear.
Insufficient (0–6)	Problem unclear, trivial, or not connected to ML.

B. Objectives & Expected Outcomes (10 points)

Level	Criteria
Excellent (9–10)	SMART objectives clearly stated; outcomes measurable and relevant.
Proficient (7–8)	Objectives are mostly clear; outcomes are stated but not well aligned.

Developing (4–6)	Objectives are general or incomplete.
Insufficient (0–3)	No clear objectives or outcomes.

C. Data Collection & Preprocessing (15 points)

Level	Criteria
Excellent (13–15)	Thorough data description; preprocessing well explained; challenges acknowledged.
Proficient (10–12)	Adequate data explanation; some preprocessing details are missing.
Developing (7–9)	Limited explanation; important preprocessing steps are missing.
Insufficient (0–6)	The data are poorly described or inappropriate for the project.

D. Methods, Models, and Technical Rigor (25 points)

Level	Criteria
Excellent (22–25)	Clear explanation of ML algorithms, model selection, hyperparameter tuning, and justification. Technical depth appropriate for a final project.
Proficient (17–21)	Good explanation of methods; some details are missing.
Developing (10–16)	Superficial coverage of models; poor justification.
Insufficient (0–9)	Methods unclear, incorrect, or missing.

E. Results, Analysis, and Visualizations (25 points)

Level	Criteria
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Excellent (23–25)	Strong results, insightful analysis, high-quality visualizations (charts, tables, confusion matrices), and clear model comparison.
Proficient (19–22)	Adequate results and analysis; visuals present but basic.
Developing (13–18)	Limited or unclear results; weak visuals.
Insufficient (0–12)	Missing results or incorrect interpretation.

F. Presentation Delivery & Professionalism (10 points)

Level	Criteria
Excellent (9–10)	Clear, engaging speaking; strong teamwork; professional slide design; stays within time.
Proficient (7–8)	Mostly clear delivery; minor timing or clarity issues.
Developing (4–6)	Uneven delivery; unclear explanations; some members under-participate.
Insufficient (0–3)	Poor delivery, incomplete participation, or unprofessional slides.